

Customer Segmentation Project Report

1. Objective

The objective of this project is to segment customers based on demographic, financial, and behavioral attributes. The analysis helps identify different customer groups and supports targeted marketing, personalized offers, and better customer relationship management.

2. Dataset Description

The dataset used for this project contains 53,503 customer records and 20 columns. It includes customer details such as age, gender, marital status, education level, geographic information, occupation, income level, behavioral data, purchase history, insurance products owned, coverage amount, premium amount, policy type, communication channel, and segmentation group.

Metric	Value
Average Age	44.14
Average Income Level	82,768.32
Average Premium Amount	3,023.70
Average Coverage Amount	492,580.79
Total Customers	53,503

3. Methodology

The dataset was imported into Power BI for cleaning, transformation, and visualization. Calculated columns were created for age group, income segment, premium segment, and coverage segment. Charts and slicers were used to analyze customer distribution and compare customer groups.

4. Segmentation Criteria

Segment Type	Description
Age Group	Young, Adult, and Senior customers based on age
Income Segment	Low, Middle, and High income customers
Premium Segment	Low, Medium, and High premium customers
Coverage Segment	Low, Medium, and High coverage customers
Segmentation Group	Existing customer segment category from the dataset

5. Dashboard Visualizations

The Power BI dashboard includes total customers, average age, average income, average premium, gender distribution, age group distribution, income segment, premium segment, policy type analysis, segmentation group analysis, geographic distribution, and an income vs premium scatter chart.

6. Key Insights

- Customer segmentation helps identify high-value customers based on income, premium amount, and coverage amount.
- Gender, age group, and geographic information help understand demographic distribution.
- Premium and coverage segments help identify customers who contribute more to business revenue.

- The income vs premium scatter chart helps compare financial capacity with premium contribution.
- Preferred communication channel analysis helps businesses contact customers through suitable channels.

7. Business Recommendations

Customer Segment	Recommended Strategy
High Premium Customers	Provide loyalty rewards, premium policy upgrades, and personalized service
Middle Income Customers	Offer affordable policy bundles and renewal discounts
Low Premium Customers	Use targeted promotions and awareness campaigns
Young Customers	Promote digital communication and flexible plans
High Coverage Customers	Provide priority support and cross-selling opportunities

8. Conclusion

This customer segmentation project successfully analyzes customer data using Power BI. The dashboard provides useful insights into customer demographics, income, premium amount, coverage amount, policy type, and customer segments. The results can help businesses improve marketing strategies, customer targeting, and decision-making.